

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

The population of the town of Smithville doubled every **75** years from **1659** to **1959**. The population of this town was **240,000** in **1959**. What was the population of this town in **1659**?

Correct Answer: 15000

Rationale

The correct answer is **15,000**. Let ***y*** represent the population of Smithville ***x*** years after **1659**. Since the population doubled every **75** years, the relationship between ***x*** and ***y*** can be represented by an exponential equation in the form **$y = a(b)^{\frac{x}{k}}$** , where ***a*** is the population of Smithville in **1659** and the population increases by a factor of ***b*** every ***k*** years. It's given that the population doubled, or increased by a factor of **2**, every **75** years. Therefore, ***b* = 2** and ***k* = 75**. Substituting **2** for ***b*** and **75** for ***k*** in the equation **$y = a(b)^{\frac{x}{k}}$** yields **$y = a(2)^{\frac{x}{75}}$** . It's also given that the population of this town was **240,000** in **1959**. The number of years from **1659** to **1959** is **1959 – 1659**, or **300** years. Therefore, when ***x* = 300**, the value of ***y*** is **240,000**. Substituting **300** for ***x*** and **240,000** for ***y*** in the equation **$y = a(2)^{\frac{x}{75}}$** yields **$240,000 = a(2)^{\frac{300}{75}}$** , or **$240,000 = a(2)^4$** , which is equivalent to **$240,000 = a(16)$** . Dividing both sides of this equation by **16** yields **$15,000 = a$** . Therefore, the population of this town in **1659** was **15,000**.